Data as a Service

High-Level Strategy and Architecture

Meeting overview

An Introduction

- **Context** - First of a series of meeting to

  GIS with, and provide transparency to, data stakeholders

  Flesh out and shape BW's Data Strategy and the DaaS architecture and implementation plans

- **Type of meeting:** This is BnL a GIS meeting that probably will feel like a "teaching one".
- **RP: GH**
- **Audience**:

  Believable / Stakeholder for the Data Strategy

  Future members of Stakeholder Group (TBD)

- **WGLL**: The audience GIS (not necessarily in agreement) with elements of the Data as a Service movie
- **Poll**: **TBD**
- **Not in Scope for <u>this</u> meeting**

  High Level 5 Step Analysis

  Life-cycle model details

  Maturity Model

  Role and Responsibilities

  Reference Architecture (Service View – DRAFT)

INTERACTION:

Groups Getting in Sync

Questions to Think About

1. General meeting prep:
   a. Be clear on the RP and goals for the meeting.
   b. Establish the agenda, meeting materials needed, people needed in the room, etc.
   c. If decisions need to be made, make sure the right people are in the room.

2. To prepare to get in sync:
   a. Make sure to crisply think about the goal of the meeting and what you need to get in sync on.
   b. Where are you on the field and how does that apply to what you need to get in sync on? Are they aware of the context to the extent they need to be?
   c. Where is the other person / group on the field and how does that apply? Do you have the info you need to? What should you be watching out for?
   d. Think about the things that you need to make sure they are aware of.
   e. What things could impact them? What dependencies exist?
   f. Think through the questions you need to ask them.
      i. What are you unclear about? What are you waiting on them for? Is there anything you need to probe?
      ii. Are there any problems you need to bring up? Any machines breaking? (Tie it to issue logs / recent threads to put a point on key observations)
      iii. Are there any plans / milestones that you need to check up on? Bring them.
   g. Review the baseball cards of who will be there.

Principles That Might Be Helpful

PREPARING

29a To avoid confusion, make clear which kind of conversation (debate, discussion, or teaching) you are having

33a Make it clear who the meeting is meant to serve and who is directing the meeting.

199 Distinguish the important things from the unimportant things and deal with the important things first.

RUNNING THE MEETING

24 Be assertive and open-minded at the same time.

24 Know when to stop debating and move on to agreeing about what should be done.

12b If it is your meeting to run, manage the conversation.

12b Enforce the logic of conversations.

33j Achieve completion in conversations.

33f Navigate the levels of the conversation clearly.

12b Understand "above the line" and "below the line" thinking and how to navigate between the two.

Introduction

Session Objectives and Summary

This is one of a series of meetings to provide transparency to data stakeholders and to iteratively shape and detail out our Strategy, Data as a Service (Daas) architecture and associated implementation plans

Objectives

- GIS on data goals, objectives and principles GIS
- on the problems and issues
- Visualize the To-Be state - WGLL
- Introduce initial elements of the DaaS Movie
  - Main Activity threads around Governance / Machinery, Platforms and Services
  - High level sequencing
- GIS on DaaS vision
- Agreement on value and need for DaaS and to move forward to Train Gate

Summary

- **Vision:** Deliver world-class data platforms and services to enable sophisticated business analytics and capabilities that forge a knowledge-based enterprise and establish long-term information advantage
- **Enterprise Architecture responsibilities**:
  - Develop overall vision and frameworks for decision making
  - Articulate principles for data life-cycle governance and management. Stand-up the essential enterprise data governance and management machines
  - Define the Services needed to support the DaaS paradigm
  - Identify and sequence the Data Platforms needed to meet business goals
  - Define the high level machinery and movie to implement all of the above
- **What we are asking:**
  - Triangulation / probing on a draft Data Strategy and DaaS model
  - Clarity on the goals and outcomes
  - Agreement on the shape of an initial Train Gate DaaS train (this approach), recognizing that the specific sub-trains will be designed individually

Document contents and description

Getting to a Data Strategy. EA's goal and initial elements for data

- The "**Data as a Service Strategy**" is developed through a combination of complementary processes that are iteratively implemented:
  - Top-down: using WGLL input from believable sources. In addition, top-down definition of a data life-cycle model and the Principles that shape WGLL
  - High level definition of an initial portfolio of services
  - Bottom-up: using the 5 steps approach for specific domain-driven analysis driven by use-cases (in progress)
- **Data is a corporate asset** managed at the enterprise level by the business and enabled by technology to:
  - Support truth-driven everything (i.e. where reality is described by accurate description of reality)
  - Connects dots throughout the organization
- **EA goals for data**: Deliver world-class data platforms and services to enable sophisticated business analytics and capabilities that forge a knowledge-based enterprise and establish long-term information advantage
  - Establish a future architecture for cloud-based data platforms and services that enable sophisticated business analytics, enterprise machines and decision making. This includes Leading the Design and Guiding the Build of leading-edge technological capabilities to process, exploit, understand and secure data
  - Accelerate accessing and the use of data by standardizing and exposing data in the form of a service
  - Foster an agile enterprise for data accessibility. Architecture for DaaS framed against the goal of loose coupling across layers. This is an important design pattern that helps keeping pace with technology in an iterative manner. Replace what you need without breaking other components
  - Establish a governance model for managing DaaS
- **This initial deck focuses on DaaS definition and goals; 4 key messages re: DaaS; and next steps**

Going back before coming forward

Reviewing, confirming and initiating the Diagnosis of data problems

Leveraged previous data and architecture analyses and triangulation with believable parties and current RP's to get a sense of the resident problems and issues with data. There is a rich body of work that is still relevant today

CURRENT STATE PROBLEMS AND ISSUES

Problem	Initial Diagnosis
Data Quality, Access and Data Usage Risks	• Lack of Data Governance and controls • Poor Data Quality • No Formal Data Architecture • Limited Security visibility
Metadata Management	• Little documentation on the meaning or definition of metadata is available • Limited metadata standards within the departments and completely lacking at the firm-wide level • No Master Data Management or other governance functions to manage metadata
Operational Risk	• Excessive Customization • Technology Obsolesce • Fragile Infrastructure • Limited Scalability • Limited skilled personnel • Key Person Risks
Security Risks	• Limited instrumentation and visibility of data and data usage • Limited security controls outside of custom applications • Gaps in audit control and historical data with difficulties tracking logs

Key message #1 – Use the "lifecycle" construct

The "big idea" and how it translates into a meta-framework

- The overriding idea is to look at a reference "lifecycle" of information and propose a series of principles regarding how data is:
  - Sourced
  - Categorized
  - Transformed
  - Used
  - Enriched
  - Curated over time
- The lifecycle is also the foundation for the definition of data-related:
  - Services
  - Machineries
  - Elements of the overall technology governance model
- As mentioned on slide 4, the approach to cover all the above is iterative with an initial focus on:
  - Top down definition of a data life-cycle model and the Principles that shape WGLL (included)
  - High level definition of an initial portfolio of services (next step)

Key message #2 – incorporate in the Xaas construct

DaaS is a component of the Reference Conceptual Architecture

Key principles:

- Each service is provided on demand
- Layer Decoupling
- Cross stack communication through enterprise bus
- Security and controls implemented through common bus with enforcement at each layer

Presentation layer providing device-agnostic support and other services (e.g. composability and persistence)

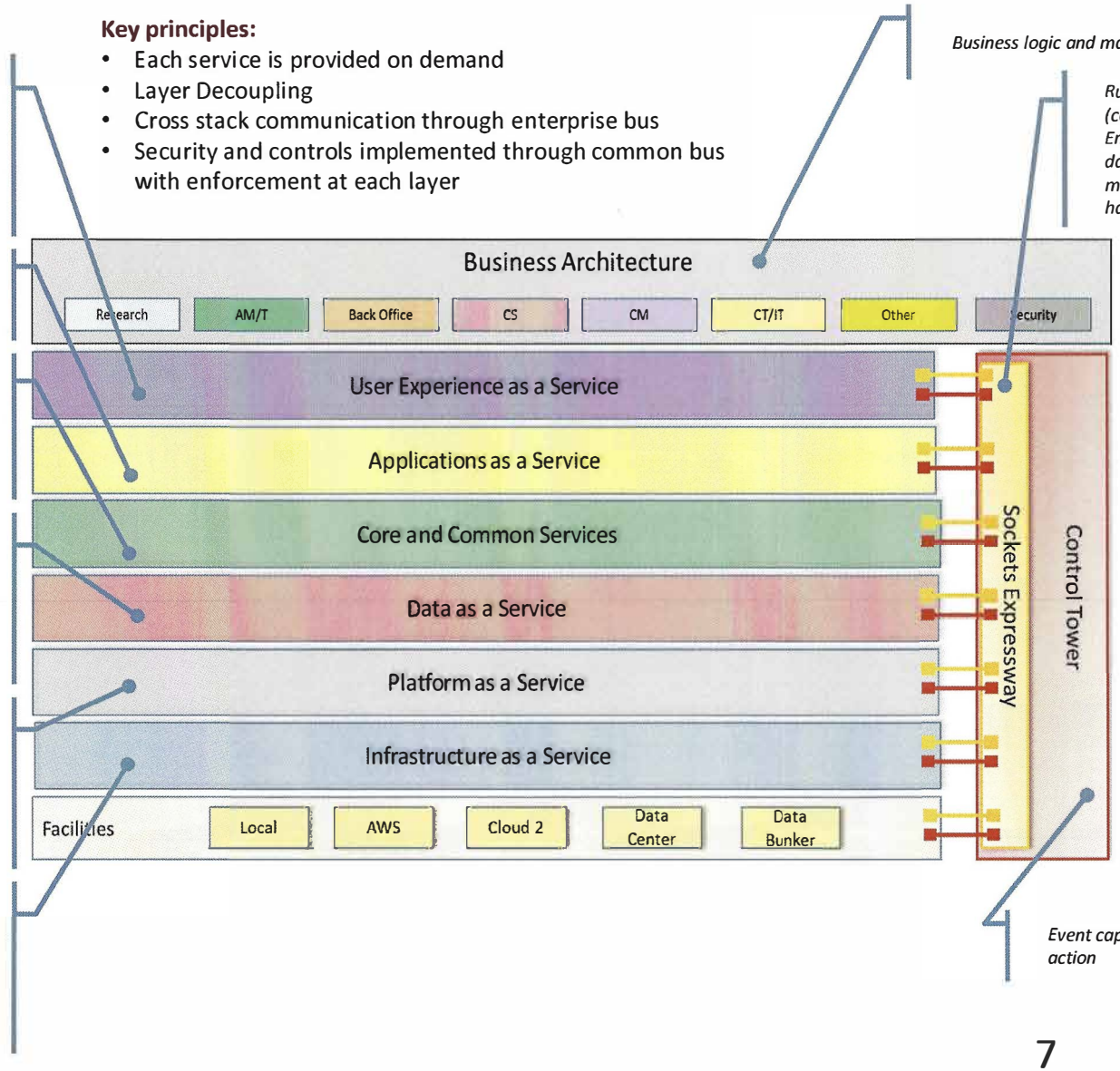
On-demand applications, managed centrally

Services shared by applications and other layers of the model (e.g. authentication, workflow...)

Data services exposed for consumption by the other layers of the stack with supporting access and encapsulation services

Defines environment to develop, deploy and host application and data services

On-demand physical or virtual infrastructure supporting the stack (Data center, telecommunications...)



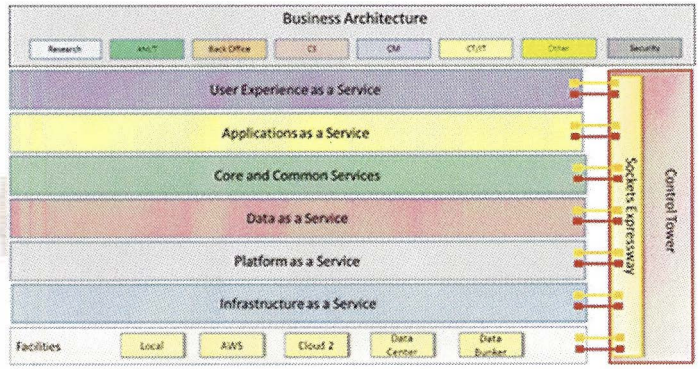
Business logic and machinery

Rule-based Multi-mode (connection/connection-less) Enterprise service bus with data transformation, mapping, orchestration, event handling ...

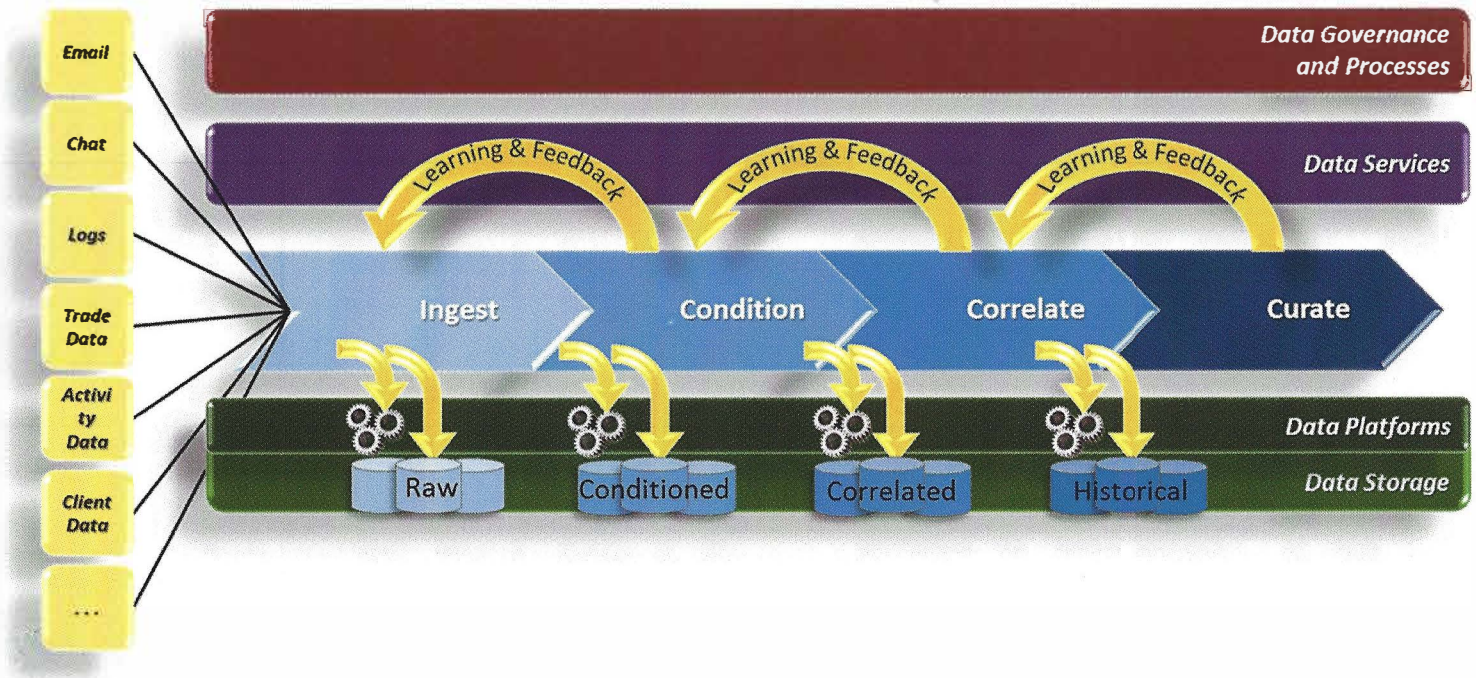
Event capture, correlation and action

Key message #2 – incorporate in the Xaas construct

DaaS is a component of the good Reference Conceptual Architecture



DaaS Architecture



Key message #3 – leverage known best practices (research and review in progress)

Big Data Best Practices from Industry

- Any data from any source can be acquired, ingested, and processed
- All data are uniquely identified, time-tagged, and classified at the time of ingestion for control and absolute traceability in space and time
- Pre-position data for performance—data is distributed into multiple engines for consumption and analysis and pre-indexed for known patterns, use models, and correlations
- Data is stored in multiple repositories and in multiple forms to meet business needs
- Design for real-time--maximize use of in-memory processing; keep data close to compute; move the smallest objects
- Multi-format data discovery – ability to navigate, search, and discover data independent of the formats
- Latency breeds contempt – data is always available for consumption in a form that does not require upfront conditioning
- Enable learning feedback to create, preserve and apply knowledge and learning through the system
- Continuous refactoring based on cumulative learning and knowledge – data is constantly reconditioned and re-correlated as knowledge evolves and learning occurs

Key message #4 – use principled thinking

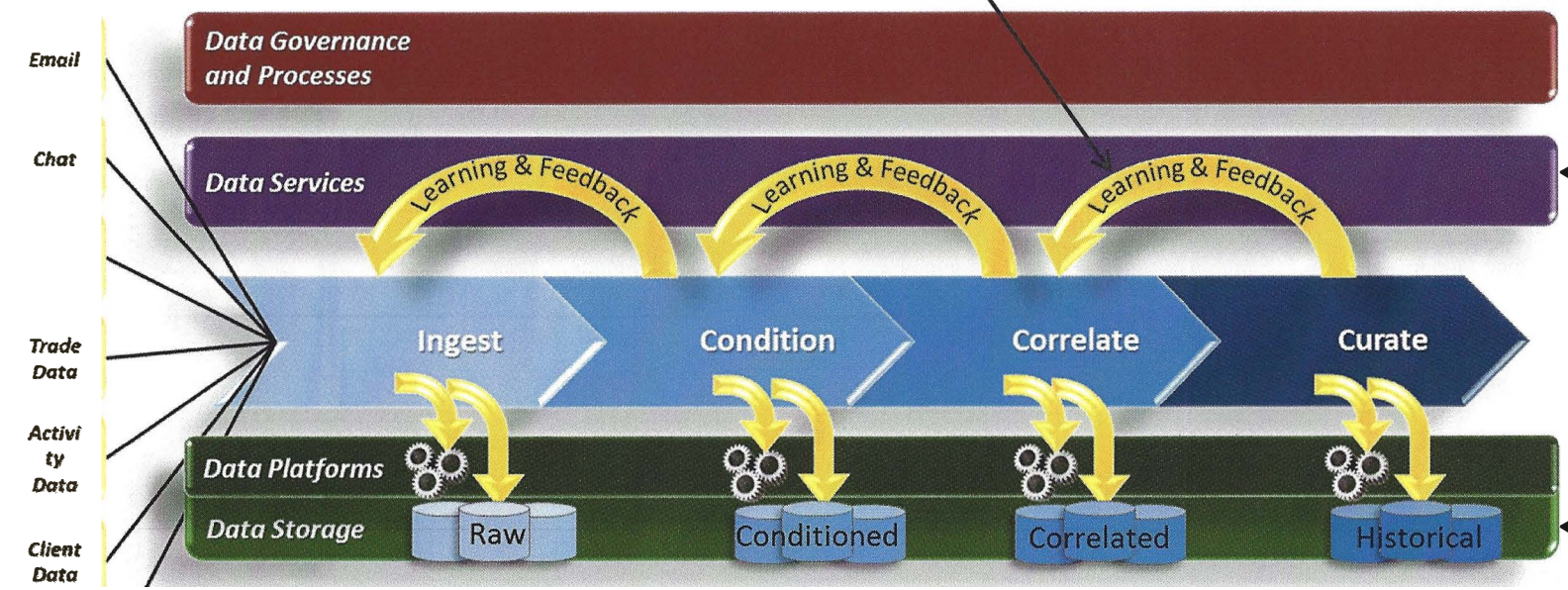
DaaS architecture high-level principles

- **Do No Harm**—Ensure that ongoing data and business activities are not adversely affected while building and standing-up the Data as a Service fabric. Leverage ongoing best practices where appropriate.
- **Build for Security** - Strong security controls are built in at the onset and designed to include monitoring functions and enable easy sockets integration to assure complete situational awareness.
- **Cloud First**—Leverage for cloud-based, Infrastructure and Platform as a Service capabilities consistent with the TD goals and objectives.
- **Loosely Coupled Systems** - A well-documented specification for communications and data interchanges between system components that facilitates the easy addition of new technologies into the system to address new business use cases.
- **Continuous Availability** - Design and build to ensure systems are always available by leveraging multi-zone, multi-cloud distribution of data, platforms, and services.
- **Everything in Real Time** - Design and build systems to maximize speed and performance, minimize latency, and operate in real-time at petabyte and beyond scales.
- **Scalability**—Implement dynamically scalable data storage and processing capabilities to sustain extreme data processing and transactional loads for complex event processing and real time analytics.
- **Agile Data Modeling**—Apply an iterative, incremental, and evolutionary approach to data modeling. Adopt best big data practices of late binding and schema on write. Implement flexible and extensible data models to support migrations, upgrades, and refactoring of data.
- **Learning Feedback**—Enable learning to be fed back into the system to build, enhance and leverage corporate knowledge.
- **Lightweight/Lean Data Governance**—Focus data governance to assure the availability, integrity, security, usability, and quality of data.
- **Standardization** – Ensure enterprise-wide standards for investment management technology and process solutions where possible relying on industry standards.
- **Data Integrity** – Apply business owned, automated procedures to ensure consistency and quality of data across the enterprise; using exception-based processes.

DaaS - pulling messages 1 – 4 together

Description and Key Concepts

Learning/Feedback loop to other steps of the lifecycle leverages automated and analysis-driven **machine learning** to **recast** previously known data on the basis cumulative knowledge obtained over time.



- Data and objects are made accessible through high-performance **services** for consumption by analytic tools and applications which perform standing and ad-hoc correlation, synthesis etc…
- Results are developed on the fly but can be stored for future references

- Each step in the lifecycle can produce/use both data and metadata
- By and large, data tends to stay immutable once produced. Some meta data might change in value and/or structure over time

- Data is ingested from external or internal sources in "**Landing Zones**" and is immediately tagged for control
- The raw data are **immutable** and stored for continuous re-ingestion in **data bunker**

- Tagged data is then **conditioned** to create **data objects** using a **Master Catalog** schema
- This is changeable set of objects whose structure can be revisited, depending on maturity of thinking, or demand …

- Conditioned objects are then **correlated across** objects and streams to discern patterns, develop hypotheses, and create connected knowledge

- Objects are preserved and stored in an application neutral, **immutable form,** frozen in time in data bunkers
- They can be retrieved and consumed into any new format

DaaS - pulling messages 1 – 4 together

Components

- **Data Governance and Processes**—data focused governance, policies, and processes directed and driven by the business and managed through enterprise architecture to control and enforce decisions on data, formats, technology, and tools that are implemented as part of Bridgewater's Data as a Service environment.

- **Data Services**—the functions and capabilities needed to operate and make the data and data platforms available for consumption by users, applications and analytics. This includes services for data ingestion, conditioning, correlation and enrichment, life cycle management, security and control, access, discovery, feedback and learning.

- **Data Platforms and Storage**—the secure IaaS, PaaS, and DaaS software and hardware products and capabilities to process, compute, store, serve, and manage data of all types. This includes platforms for: NoSQL, SQL, Streams, Hadoop/Map Reduce, Full Text, Graph, Semantic, Video, Audio, and other processing needs.

DaaS - pulling messages 1 – 4 together

Description and Key Concepts

- **Ingestion**: Data of any type is ingested from external or internal sources to "Landing Zones" where it is immediately tagged, classified, processed and marked for access control and absolute traceability in space and time. The raw data are immutable. Once acquired and processed a copy of the original data plus the associated controlling metadata is stored for access, replay, or re-ingestion as needed. Data is automatically placed into multiple platforms and storage environments—NoSQL, SQL, Hadoop/Map Reduce, etc—as appropriate to the data type and to enable multiple tools and applications to immediately make use of it.

- **Conditioning**: Conditioning applies specific transformations-marking and tagging, normalization, master data controls, extracted/identified entities, etc—to the data objects and data streams. Conditioned data is changeable as tools, technologies, insights, experience, and business needs change and mature. Conditioned data should be regularly revisited and updated and refined as needed. Conditioned data is indexed and automatically placed into multiple platforms and storage environments for immediate consumption by business users and tools for exploitation and to apply local analytics and transforms as needed.

- **Correlation**: Correlated data has connected and linked data across the objects and streams to create an integrated picture of information. Multiple views of the integrated pictures are created based upon the business area and analytic requirements. Correlated data is constantly changing and highly dynamic as new questions get asked, events occur, learning matures, or new business needs emerge.

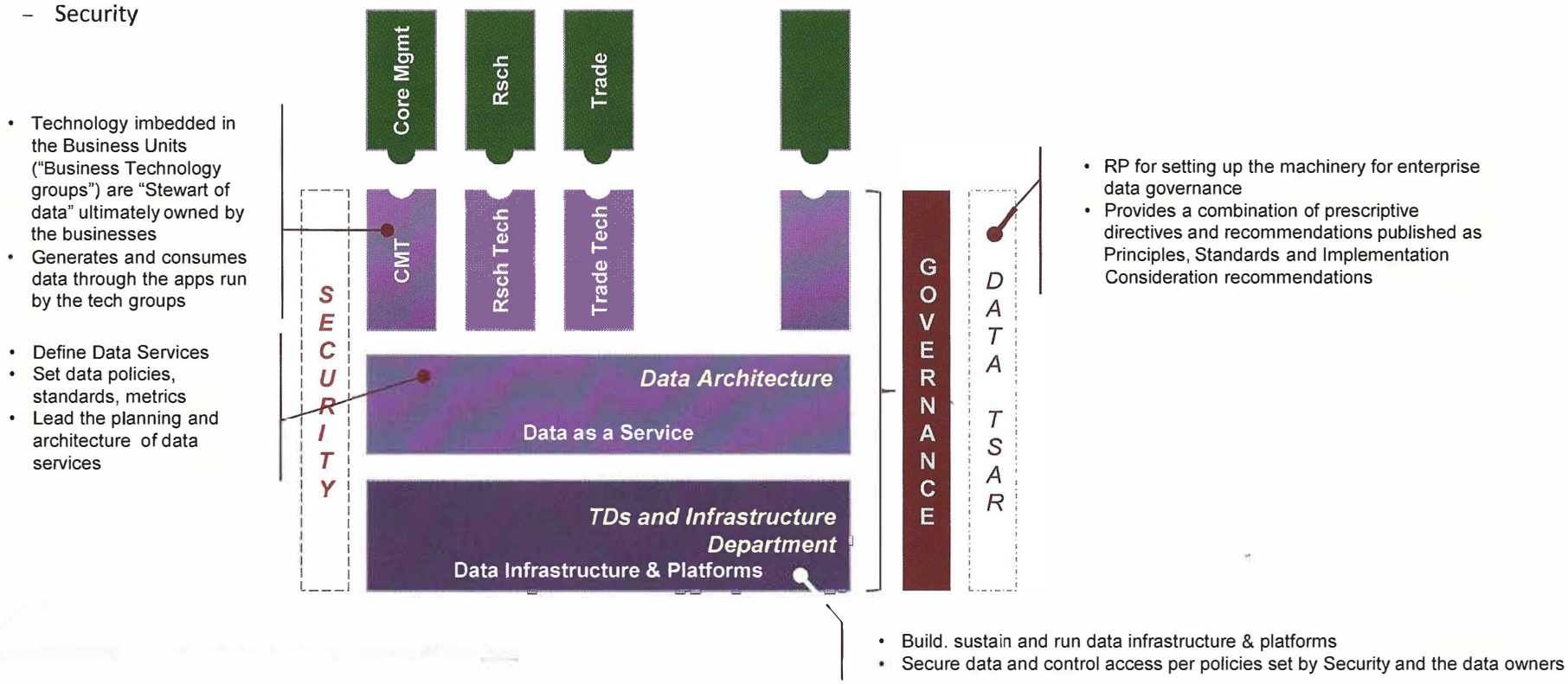
- **Curation**: Services to ensure the preservation of historic, business critical information in an immutable, standards-based, application-neutral form. Data are stored in the Data Bunker and form the fondation data store whereby time-based truths can be leveraged to inform current analyses. Foundation Data content is fixed and immutable and, to the maximum extent possible, Foundation Data formats do not change. Foundation Data services change to ensure readability as software application formats change so that Foundation Data remains constantly accessible and usable.

- **Learning/Feedback Loops**: The entire process is constantly updated and re-informed through learning feedback loops. Automated and analysis driven machine learning from the steps ahead are continuously recast through the system in order to capture knowledge and maximize data value as early as possible in the process.

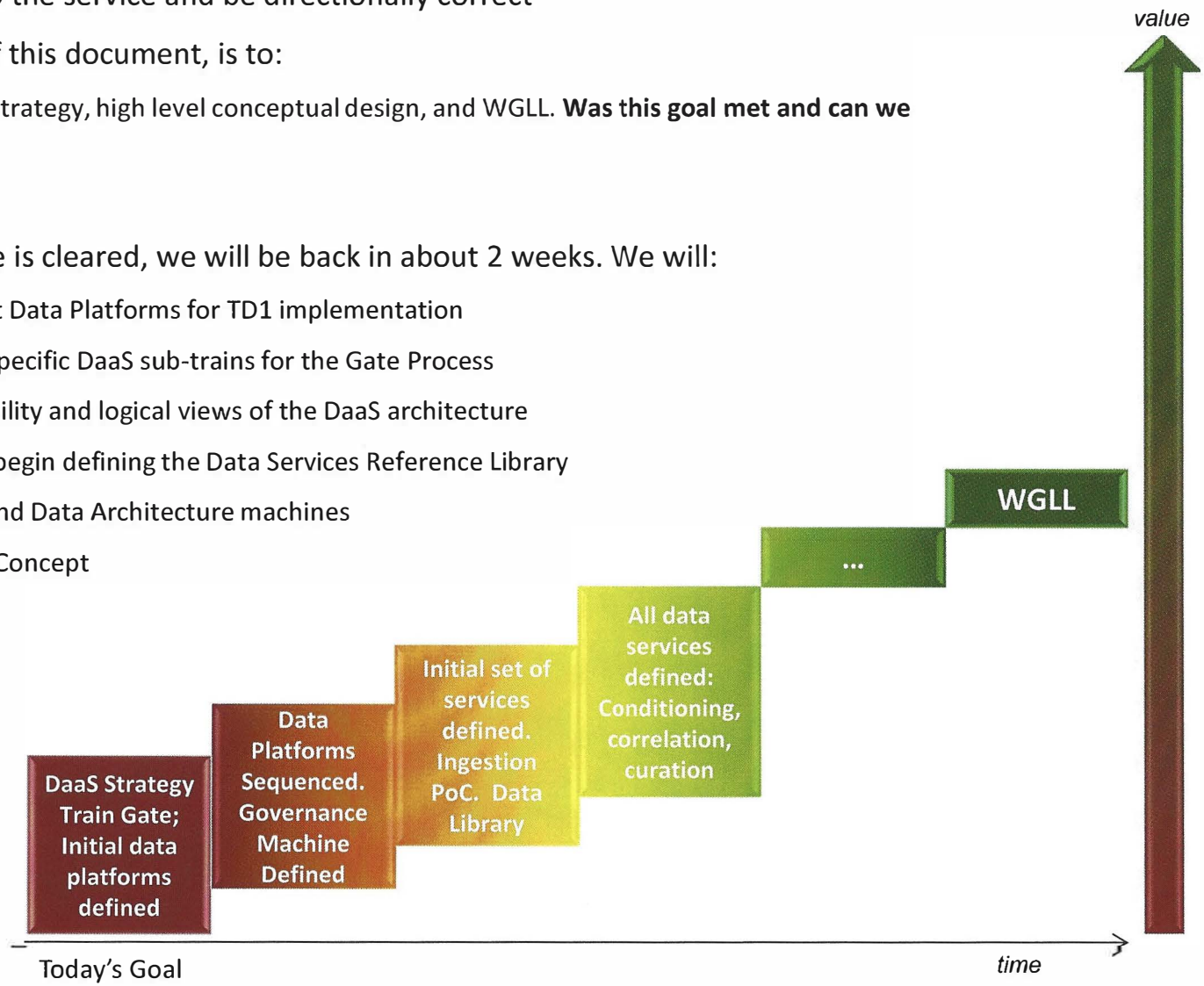
Key Message #5 – it's the who: it will take a village

Data Ecosystem: Who Does What

- Information/Data is an asset that is touched directly or indirectly by a number of constituents across the firm – Because of its strategic importance, it might ultimately be warrant the assignment of an overall RP ("Data Tsar") to implement an overall machinery to manage data governance and the data lifecycle (and perhaps an entire Technical Directive).
- As a first loop, governance will be assumed at the Enterprise Architecture level (with a dedicated Data domain architecture) with a focus on defining the Data as a Service architecture layer
- Within the "Technology Realm" the groups involved include:
  - The technology groups imbedded in the Business Units
  - The Enterprise Architecture group
  - The Infrastructure department
  - Security



DaaS High Level Movie/Roadmap and Next Steps

- To implement DaaS, we propose a series of very well defined plateaus, each delivering tangible, incremental improvement to the service and be directionally correct
- The first plateau, and goal of this document, is to:
  - GIS on the overarching DaaS strategy, high level conceptual design, and WGLL. **Was this goal met and can we proceed to the Train Gate?**

- Next steps – if the Train Gate is cleared, we will be back in about 2 weeks. We will:
  - Identify and sequence the next Data Platforms for TD1 implementation
  - Define the next plateaus and specific DaaS sub-trains for the Gate Process
  - Develop and publish the capability and logical views of the DaaS architecture
  - From the logical architecture, begin defining the Data Services Reference Library
  - Design the Data Governance and Data Architecture machines
  - Identify options for a Proof of Concept